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Driving substrate, method for fabricating same, and display panel

Abstract

A driving substrate includes a substrate, a source electrode, a drain electrode, an active layer, and a gate electrode. The source electrode and the drain electrode are disposed on the substrate. There is a first space between the source electrode and the drain electrode. The active layer is disposed on the substrate, the source electrode, and the drain electrode, and includes a channel, a source region, and a drain region. The source region covers at least part of the source electrode. The drain region covers at least part of the drain electrode. The channel corresponds to the first space. The gate electrode is disposed on a side of the active layer away from the source electrode and the drain electrode, and is disposed corresponding to the active layer. A method is used for fabricating the driving substrate. A display panel includes the driving substrate.

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Background/Summary

FIELD OF INVENTION

(1) The present disclosure relates to the technical field of display, and particularly to a driving substrate, a method for fabricating the same, and a display panel.

BACKGROUND

(2) Thin film transistors (TFTs) have been widely used in driving of active displays. As a size of a display continues to increase, so does its driving frequency. For a thin film transistor, an electron mobility of a channel region of an active layer affects an operating frequency of the thin film transistor. Improvement of the electron mobility is beneficial to improvement of performances of the thin film transistor.

(3) In order to improve the electron mobility of the thin film transistor, a material of the active layer may be changed or a width of the channel region of the thin film transistor may be increased. However, an increase in the width of the channel region of the thin film transistor leads to an increase in a size of an entirety of the thin film transistor, which is not conducive to realizing a high-resolution display. As a result, the improvement of the electron mobility of the thin film transistor is limited.

SUMMARY OF DISCLOSURE

(4) The present disclosure provides a driving substrate, a method for fabricating the same, and a display panel to solve a technical problem that improvement of an electron mobility of a current thin film transistor is limited.

(5) In order to solve the above technical problem, the present disclosure provides the following technical solutions.

(6) The present disclosure provides a driving substrate comprising a substrate, a source electrode, a drain electrode, an active layer, and a gate electrode. The source electrode and the drain electrode are disposed on the substrate. There is a first space between the source electrode and the drain electrode. The active layer is disposed on the substrate, the source electrode, and the drain electrode, and comprises a channel and a source region and a drain region on both sides of the channel. The source region covers at least part of the source electrode. The drain region covers at least part of the drain electrode. The channel corresponds to the first space. The gate electrode is disposed on a side of the active layer away from the source electrode and the drain electrode, and is disposed corresponding to the active layer.

(7) In an embodiment, an orthographic projection of the gate electrode on the substrate covers an orthographic projection of the channel of the active layer on the substrate, covers an orthographic projection of at least part of the source region of the active layer on the substrate, and covers an orthographic projection of at least part of the drain region of the active layer on the substrate.

(8) In an embodiment, the driving substrate further comprises an auxiliary source electrode and an auxiliary drain electrode. The auxiliary source electrode is disposed between the source electrode and the substrate and is electrically connected to the source electrode. The auxiliary drain electrode is disposed between the drain electrode and the substrate and is electrically connected to the drain electrode.

(9) In an embodiment, the driving substrate further comprises a conductive layer disposed on a side of the gate electrode away from the active layer, insulated from the gate electrode, and comprising a first electrode and a connecting wire that are spaced apart. The first electrode is electrically connected to the source electrode and the auxiliary source electrode, and the connecting wire is electrically connected to the drain electrode and the auxiliary drain electrode.

(10) In an embodiment, an orthographic projection of the first electrode on the substrate at least covers the orthographic projection of the channel on the substrate.

(11) In an embodiment, the driving substrate further comprises a buffer layer covering the auxiliary source electrode and the auxiliary drain electrode. The source electrode and the drain electrode are disposed on the buffer layer, and are respectively electrically connected to the corresponding source

auxiliary electrode and the drain auxiliary electrode through via holes of the buffer layer.

(12) In an embodiment, in a direction perpendicular to the substrate, a thickness of the source electrode is less than a thickness of the auxiliary source electrode, and a thickness of the drain electrode is less than a thickness of the auxiliary drain electrode.

(13) In an embodiment, an orthographic projection of the auxiliary source electrode on the substrate at least covers the orthographic projection of the channel on the substrate.

(14) In an embodiment, the driving substrate further comprises an auxiliary source electrode and an auxiliary drain electrode that are disposed on a side of the gate electrode away from the active layer and are insulated from the gate electrode. The auxiliary source electrode is electrically connected to the source electrode, and the auxiliary drain electrode is electrically connected to the drain electrode.

(15) In an embodiment, the source electrode and the drain electrode are made of metal oxide.

(16) The present disclosure further provides a display panel comprising the driving substrate of one of the foregoing embodiments.

(17) The present disclosure further provides a method for fabricating a driving substrate, comprising: providing a substrate; forming a source electrode and a drain electrode on the substrate, wherein there is a first space between the source electrode and the drain electrode; forming an active layer on the substrate, the source electrode, and the drain electrode, wherein the active layer comprises a channel and a source region and a drain region on both sides of the channel, the source region covers at least part of the source electrode, the drain region covers at least part of the drain electrode, and the channel corresponds to the first space; and forming a gate electrode on a side of the active layer away from the substrate, wherein an orthographic projection of the gate electrode on the substrate covers an orthographic projection of the channel of the active layer on the substrate, covers an orthographic projection of at least part of the source region of the active layer on the substrate, and covers an orthographic projection of at least part of the drain region of the active layer on the substrate.

(18) In an embodiment, before the forming the source electrode and the drain electrode on the substrate, the method further comprises: forming a first metal thin film on the substrate by physical vapor deposition; patterning the first metal thin film to form an auxiliary source electrode and an auxiliary drain electrode; and depositing a buffer layer on the auxiliary source electrode, the auxiliary drain electrode, and the substrate by chemical vapor deposition; wherein the source electrode and the drain electrode are formed on the buffer layer.

(19) In an embodiment, the method further comprises: forming a conductive layer on a side of the gate electrode away from the active layer, wherein the conductive layer comprises a first electrode and a connecting wire that are spaced apart, the first electrode is electrically connected to the source electrode and the auxiliary source electrode, and the connecting wire is electrically connected to the drain electrode and the auxiliary drain electrode.

(20) The present disclosure provides a driving substrate, a method for fabricating the same, and a display panel. The driving substrate comprises a substrate, a source electrode, a drain electrode, and an active layer. The source electrode, the drain electrode, and the active layer are disposed on the substrate. There is a first space between the source electrode and the drain electrode. The active layer is further disposed on the source electrode and the drain electrode. The active layer comprises a channel and a source region and a drain region on both sides of the channel. The source region covers at least part of the source electrode. The drain region covers at least part of the drain electrode. The channel corresponds to the first space. In this way, a length of the channel of the active layer is defined by a space between the source electrode and the drain electrode, so as to reduce the length of the channel of the active layer, thereby improving an electron mobility of the channel of the active layer and solving a problem that improvement of an electron mobility of a current thin film transistor is limited.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) In order to more clearly illustrate technical solutions in embodiments or the prior art, a brief description of accompanying drawings used in the embodiments or the prior art will be given below. The accompanying drawings in the following description are merely some embodiments of the present disclosure. For those skilled in the art, other drawings may be obtained from these accompanying drawings without creative labor.
- (2) FIG. 1 is a schematic cross-sectional structure diagram of a driving substrate according to an embodiment of the present disclosure.
- (3) FIG. 2 is another schematic cross-sectional structure diagram of a driving substrate according to an embodiment of the present disclosure.
- (4) FIG. 3 is yet another schematic cross-sectional structure diagram of a driving substrate according to an embodiment of the present disclosure.
- (5) FIG. 4 is a schematic flowchart of a method for fabricating a driving substrate according to an embodiment of the present disclosure.
- (6) FIG. 5 is a schematic cross-sectional structure diagram of a substrate according to an embodiment of the present disclosure.
- (7) FIG. 6 is a schematic cross-sectional structure diagram of forming an auxiliary source electrode and an auxiliary drain electrode on the substrate of FIG. 5.
- (8) FIG. 7 is a schematic cross-sectional structure diagram of forming a source electrode and a drain electrode on a structure of FIG. 6.
- (9) FIG. 8 is a schematic cross-sectional structure diagram of forming an active layer on a structure of FIG. 7.
- (10) FIG. 9 is a schematic cross-sectional structure diagram of forming a gate electrode on a structure of FIG. 8.
- (11) FIG. 10 is a schematic cross-sectional structure diagram of forming a passivation layer on a structure of FIG. 9.

DETAILED DESCRIPTION

- (12) The following description of various embodiments of the present disclosure with reference to the accompanying drawings is used to illustrate specific embodiments that can be practiced. Directional terms mentioned in the present disclosure, such as “above”, “below”, “front”, “back”, “left”, “right”, “inside”, “outside”, “side”, are merely used to indicate the direction of the accompanying drawings. Therefore, the directional terms are used for illustrating and understanding the present disclosure rather than limiting the present disclosure. In the figures, elements with similar structure are indicated by the same reference numerals. In the accompanying drawings, thicknesses of some layers and regions are exaggerated for clarity of understanding and ease of description. That is, a size and thickness of each component shown in the accompanying drawings are arbitrarily shown, and thus the present application is not limited thereto.
- (13) Please refer to FIG. 1, which is a schematic cross-sectional structure diagram of a driving substrate **100** according to an embodiment of the present disclosure. The driving substrate **100** comprises a substrate **10**, a source electrode **20**, a drain electrode **30**, an active layer **40**, and a gate electrode **50**. The source electrode **20**, the drain electrode **30**, the active layer **40**, and the gate electrode **50** are disposed on the substrate **10**. There is a first space between the source electrode **20** and the drain electrode **30**. The active layer **40** is disposed on the substrate **10**, the source electrode **20**, and the drain electrode **30**. The active layer **40** comprises a channel **41** and a source region **42** and a drain region **43** on both sides of the channel **41**. The source region **42** covers at least part of the source electrode **20**. The drain region covers **43** at least part of the drain electrode **30**. The channel **41** corresponds to the first space. The gate electrode **50** is disposed on a side of the active

layer **40** away from the source electrode **20** and the drain electrode **30**, and is disposed corresponding to the active layer **40**.

(14) In this embodiment, a length of the channel **41** of the active layer **40** is defined by the first space between the source electrode **20** and the drain electrode **30**, so as to achieve a shorter length of the channel **41**. In this way, the length of the channel **41** of the active layer **40** is reduced, thereby increasing an electron mobility of the channel **41** of the active layer **40**, and solving a problem that improvement of an electron mobility of a current thin film transistor is limited.

(15) In an embodiment, the driving substrate **100** further comprises an auxiliary source electrode **21** and an auxiliary drain electrode **31**. The auxiliary source electrode **21** is disposed between the source electrode **20** and the substrate **10** and is electrically connected to the source electrode **20**, so as to reduce a resistance of the source electrode **20**. The auxiliary drain electrode **31** is disposed between the drain electrode **30** and the substrate **10** and is electrically connected to the drain electrode **30**, so as to reduce a resistance of the drain electrode **30**.

(16) Optionally, the substrate **10** may be a rigid substrate or a flexible substrate. When the substrate **10** is a rigid substrate, the substrate **10** may be a glass substrate, a plastic substrate, or other rigid substrate. When the substrate **10** is a flexible substrate, the substrate **10** may be a polyimide (PI) film, an ultra-thin glass film, or other flexible substrate.

(17) The auxiliary source electrode **21** and the auxiliary drain electrode **31** are disposed on the substrate **10**. There is a space between the auxiliary source electrode **21** and the auxiliary drain electrode **31**. The auxiliary source electrode **21** and the auxiliary drain electrode **31** are insulated from each other. Optionally, the auxiliary source electrode **21** and the auxiliary drain electrode **31** are made of one or a combination of low resistivity metals such as Mo, Al, Cu, and Ti, for example, Mo/Al, Mo/Cu, MoTi/Cu, MoTi/Cu/MoTi, Ti/Al/Ti, Ti/Cu/Ti, Mo/Cu/IZO, IZO/Cu/IZO, or Mo/Cu/ITO.

(18) The driving substrate **100** further comprises a buffer layer **11** covering the auxiliary source electrode **21** and the auxiliary drain electrode **31**. The source electrode **20** and the drain electrode **30** are disposed on a side of the buffer layer **11** away from the substrate **10**. The active layer **40** covers at least part of the source electrode **20**, at least part of the drain electrode **30**, and the buffer layer **11**.

(19) Specifically, the buffer layer **11** covers the auxiliary source electrode **21**, the auxiliary drain electrode **31**, and the substrate **10**. The buffer layer **11** prevents undesired impurities or contaminants (e.g., moisture, oxygen, etc.) from diffusing from the substrate **10** into a device that may be damaged by these impurities or contaminants. Optionally, the buffer layer **11** is made of silicon oxide (SiO_x), silicon nitride (SiN_x), or a stack of silicon oxide and silicon nitride.

(20) Meanwhile, the buffer layer **11** further provides a flat top surface, so as to facilitate formation of the source electrode **20**, the drain electrode **30**, and the active layer **40** on the buffer layer **11**. The flat top surface is more favorable for the active layer **40** to have good interface properties.

(21) The source electrode **20** and the drain electrode **30** are formed on the buffer layer **11**. There is the first space between the source electrode **20** and the drain electrode **30**. The source electrode **20** and the drain electrode **30** are made ITO, IZO, AnCl, or other metal oxide with low resistivity. For example, the source electrode **20** and the drain electrode **30** are made of a metal oxide such as ITO. A difference in an etching size of a metal oxide such as ITO in a yellow light process is small. Due to a limitation of an exposure machine, a length of the first space between the source electrode **20** and the drain electrode **30** is made smaller, so that the active layer **40** has one channel **41** with a smaller length, thereby improving the electron mobility of the channel **41** of the active layer **40**.

(22) Furthermore, in a direction perpendicular to the substrate **10** and from the substrate **10** to the source electrode **20**, a thickness of the auxiliary source electrode **21** is greater than a thickness of the source electrode **20**, and a thickness of the auxiliary drain electrode **31** is greater than a thickness of the drain electrode **30**, so as to further reduce a resistance of the auxiliary source electrode **21** and a resistance of the auxiliary drain electrode **31**.

(23) The active layer **40** covers at least part of the source electrode **20**, at least part of the drain electrode **30**, and the buffer layer **11**. The active layer **40** is formed on the source electrode **20** and the drain electrode **30** to prevent the active layer **40** from being affected by an etching solution when the source electrode **20** and the drain electrode **30** are formed by etching. If the etching solution is in contact with the active layer **40**, the etching solution will etch the active layer **40** and change In, Ga, and Zn plasma concentrations on a surface of the active layer **40**, thereby affecting stability and reliability of the device. Optionally, the active layer **40** is made of a semiconductor metal oxide such as IGZO, IGZTO, IGTO, IGO, or AZTO.

(24) The channel **41** of the active layer **40** is disposed corresponding to the first space. In the direction perpendicular to the substrate **10** and from the substrate **10** to the source electrode **20**, a thickness of the active layer **40** is greater than the thickness of the source electrode **20** or the thickness of the drain electrode **30**. Therefore, the channel **41** of the active layer **40** is located in the first space, and a height of the channel **41** is greater than a height of the source electrode **20** and the drain electrode **30**. The source region **42** of the active layer **40** covers the source electrode **20**, and the drain region **43** of the active layer **40** covers the drain electrode **30**. Accordingly, surfaces of the source region **42**, the channel **41**, and the drain region **43** of the active layer **40** away from the buffer layer **11** are horizontal, so that a surface of the active layer **40** away from the buffer layer **11** is horizontal.

(25) An orthographic projection of the channel **41** of the active layer **40** on the substrate **10** falls within an orthographic projection of the auxiliary source electrode **21** on the substrate **10**. Therefore, the auxiliary source electrode **21** further has a light shielding function to prevent light from irradiating the channel **41** of the active layer **40**. Moreover, the orthographic projection of the auxiliary source electrode **21** on the substrate **10** may further cover an orthographic projection of an entirety of the active layer **40** on the substrate **10**. That is, orthographic projections of the source region **42** and the drain region **43** of the active layer **40** on the substrate **10** also fall within the orthographic projection of the auxiliary source electrode **21** on the substrate **10**.

(26) Furthermore, an orthographic projection of the source electrode **20** on the substrate **10** falls within the orthographic projection of the auxiliary source electrode **21** on the substrate **10**. An area of the orthographic projection of the source electrode **20** on the substrate **10** is less than an area of the orthographic projection of the auxiliary source electrode **21** on the substrate **10**. Meanwhile, an area of an orthographic projection of the drain electrode **30** on the substrate **10** is less than an area of an orthographic projection of the auxiliary drain electrode **31** on the substrate **10**.

(27) How to electrically connect the source electrode **20** and the drain electrode **30** to the auxiliary source electrode **21** and the auxiliary drain electrode **31** respectively will be described in detail below.

(28) In an embodiment, the gate electrode **50** is located on the active layer **40** and is insulated from the active layer **40**. The gate electrode **50** is disposed corresponding to the active layer **40**. Specifically, the driving substrate **100** further comprises a gate insulating layer **12** disposed between the gate electrode **50** and the active layer **40**. The gate insulating layer **12** covers the active layer **40**. The gate electrode **50** is disposed on the gate insulating layer **12**.

(29) The gate insulating layer **12** is also disposed corresponding to the active layer **40**. An orthographic projection of the gate insulating layer **12** on the substrate **10** overlaps the orthographic projection of the active layer **40** on the substrate **10**. Optionally, the gate insulating layer **12** is made of one or a combination of SiOx, SiNx, and Al2O3, for example, Al2O3/SiNx/SiOx, or SiOx/SiNx/SiOx.

(30) The gate electrode **50** is disposed on the gate insulating layer **12**. Optionally, a material of the gate electrode **50** is same as a material of the auxiliary source electrode **21**. An orthographic projection of the gate electrode **50** on the substrate **10** covers an orthographic projection of the channel **41** of the active layer **40** on the substrate **10**, covers an orthographic projection of at least part of the source region **42** of the active layer **40** on the substrate **10**, and covers an orthographic

projection of at least part of the drain region **43** of the active layer **40** on the substrate **10**.

Therefore, in the direction perpendicular to the substrate **10**, the gate electrode **50** partially overlaps the source electrode **20**, and the gate electrode **50** partially overlaps the drain electrode **30**.

(31) It can be understood that when there is a driving voltage on the gate electrode **50**, an electric field is formed between the gate electrode **50** and the corresponding source electrode **20**, and such electric field can change the source region **42** of the active layer **40** between the gate electrode **50** and the source electrode **20** from a semiconductor to a conductor. Similarly, when there is a driving voltage on the gate electrode **50**, an electric field is formed between the gate electrode **50** and the corresponding drain electrode **30**, and such electric field can change the drain region **43** of the active layer **40** between the gate electrode **50** and the drain electrode **30** from a semiconductor to a conductor. Therefore, the source region **42** and the drain region **43** of the active layer **40** do not need to be conductorized by a conductorization process, such as plasma doping.

(32) In an embodiment, the driving substrate **100** further comprises a conductive layer **60** disposed on a side of the gate electrode **50** away from the active layer **40** and insulated from the gate electrode **50**. The conductive layer **60** comprises a first electrode **61** and a connecting wire **62** that are spaced apart. The first electrode **61** is electrically connected to the source electrode **20** and the auxiliary source electrode **21**, so as to realize electrical connection between the source electrode **20** and the auxiliary source electrode **21**, and simultaneously realize electrical connection between the first electrode **61** and the source electrode **20**. The connecting wire **62** is electrically connected to the drain electrode **30** and the auxiliary drain electrode **31**, so as to realize electrical connection between the drain electrode **30** and the auxiliary drain electrode **31**.

(33) Optionally, the driving substrate **100** further comprises a passivation layer **13** disposed between the conductive layer **60** and the gate electrode **50**. The passivation layer **13** covers the gate electrode **50**, a part of the source electrode **20**, a part of the drain electrode **30**, and the buffer layer **11**. A plurality of first via holes **131** and a plurality of second via holes **132** are formed in the passivation layer **13**. The first via holes **131** penetrate through the passivation layer **13**, one of the first via holes **131** exposes another part of the source electrode **20**, and the other one of the first via holes **131** exposes another part of the drain electrode **30**. The second via holes **132** penetrate through the passivation layer **13** and the buffer layer **11**, one of the second via holes **132** exposes a part of the auxiliary source electrode **21**, and the other one of the second via holes **132** exposes a part of the auxiliary drain electrode **31**.

(34) The first electrode **61** is electrically connected to the source electrode **20** through one first via hole **131**. At the same time, the first electrode **61** is also electrically connected to the auxiliary source electrode **21** through one second via hole **132**. The connecting wire **62** is electrically connected to the drain electrode **30** through the other first via hole **131**. At the same time, the connecting wire **62** is also electrically connected to the auxiliary drain electrode **31** through the other second via hole **132**. In this way, a mask can realize the electrical connection between the source electrode **20** and the corresponding auxiliary source electrode **21**, the electrical connection between the drain electrode **30** and the corresponding auxiliary drain electrode **31**, and the electrical connection between the first electrode **61** and the source electrode **20**, which is beneficial to save cost.

(35) In an embodiment, an orthographic projection of the first electrode **61** on the substrate **10** at least covers the orthographic projection of the channel **41** on the substrate **10**. The first electrode **61** is set to have a larger area, which can prevent moisture in an environment from diffusing to the channel **41** of the active layer **40**, thereby improving the reliability of the device.

(36) Optionally, in order to better block the moisture in the environment, the orthographic projection of the first electrode **61** on the substrate **10** may cover an orthographic projection of an entirety of the gate electrode **50** on the substrate **10**, so that to form a larger area of the first electrode **61** has a larger area.

(37) In an embodiment, please refer to FIG. 2, which is another schematic cross-sectional structure

diagram of a driving substrate **101** according to an embodiment of the present disclosure. Different from the aforementioned embodiment, in the driving substrate **101**, a source electrode **20** and a drain electrode **30** are respectively electrically connected to a corresponding auxiliary source electrode **21** and a corresponding auxiliary drain electrode **31** through via holes of a buffer layer **11**. (38) Specifically, the buffer layer **11** is provided with a third via hole **111** and a fourth via hole **112**. The third via hole **111** penetrates through the buffer layer **11** and exposes a part of the auxiliary source electrode **21**. The fourth via hole **112** penetrates through the buffer layer **11** and exposes a part of the auxiliary drain electrode **31**. The source electrode **20** is electrically connected to the auxiliary source electrode **21** through the third via hole **111**. The drain electrode **30** is electrically connected to the auxiliary drain electrode **31** through the fourth via hole **112**. For other descriptions, please refer to the aforementioned embodiment, which will not be described in detail herein.

(39) In an embodiment, please refer to FIG. 3, which is yet another schematic cross-sectional structure diagram of a driving substrate **102** according to an embodiment of the present disclosure. Different from the aforementioned embodiments, in the driving substrate **102**, an auxiliary source electrode **21** and an auxiliary drain electrode **31** are disposed on a side of a gate electrode **50** away from an active layer **40** and are insulated from the gate electrode **50**, the auxiliary source electrode **21** is electrically connected to a source electrode **20**, and the auxiliary drain electrode **31** is electrically connected to a drain electrode **30**.

(40) Specifically, the auxiliary source electrode **21** and the auxiliary drain electrode **31** are disposed on a passivation layer **13**. The passivation layer **13** is provided with a plurality of via holes penetrating the passivation layer **13** and exposing a part of the source electrode **20** and a part of the drain electrode **30**. The auxiliary source electrode **21** is electrically connected to the source electrode **20** through one via hole in the passivation layer **13**. The auxiliary drain electrode **31** is electrically connected to the drain electrode **30** through the other via hole in the passivation layer **13**.

(41) The driving substrate **102** further comprises a planarization layer **14** covering the auxiliary source electrode **21** and the auxiliary drain electrode **31**. A first electrode **61** is disposed on the planarization layer **14**. The planarization layer **14** is provided with a via hole penetrating the planarization layer **14** and exposing a part of the auxiliary source electrode **21**. The first electrode **61** is electrically connected to the auxiliary source electrode **21** through the via hole in the planarization layer **14**. For other descriptions, please refer to the aforementioned embodiments, which will not be described in detail herein.

(42) The present disclosure further provides a method for fabricating a driving substrate. Please refer to FIG. 4 to FIG. 10. FIG. 4 is a schematic flowchart of a method for fabricating a driving substrate according to an embodiment of the present disclosure. FIG. 5 is a schematic cross-sectional structure diagram of a substrate according to an embodiment of the present disclosure. FIG. 6 is a schematic cross-sectional structure diagram of forming an auxiliary source electrode and an auxiliary drain electrode on the substrate of FIG. 5. FIG. 7 is a schematic cross-sectional structure diagram of forming a source electrode and a drain electrode on a structure of FIG. 6. FIG. 8 is a schematic cross-sectional structure diagram of forming an active layer on a structure of FIG. 7. FIG. 9 is a schematic cross-sectional structure diagram of forming a gate electrode on a structure of FIG. 8. FIG. 10 is a schematic cross-sectional structure diagram of forming a passivation layer on a structure of FIG. 9. The method for fabricating the driving substrate comprises the following steps.

(43) **S301**: providing a substrate **10**.

(44) Specifically, the substrate **10** is provided, as shown in FIG. 5. The substrate **10** may be a rigid substrate or a flexible substrate. When the substrate **10** is a rigid substrate, the substrate **10** may be a glass substrate, a plastic substrate, or other rigid substrate. When the substrate **10** is a flexible substrate, the substrate **10** may be a polyimide (PI) film, an ultra-thin glass film, or other flexible

substrate.

(45) **S302**: forming a source electrode **20** and a drain electrode **30** on the substrate **10**, wherein there is a first space between the source electrode **20** and the drain electrode **30**.

(46) Specifically, a first metal thin film is formed on the substrate **10** by a deposition process such as physical vapor deposition, and the first metal thin film is patterned to form an auxiliary source electrode **21** and an auxiliary drain electrode **31**, as shown in FIG. **6**. Optionally, the first metal thin film is made of one or a combination of low resistivity metals such as Mo, Al, Cu, and Ti, for example, Mo/Al, Mo/Cu, MoTi/Cu, MoTi/Cu/MoTi, Ti/Al/Ti, Ti/Cu/Ti, Mo/Cu/IZO, IZO/Cu/IZO, or Mo/Cu/ITO.

(47) Then, a buffer layer **11** is deposited on the auxiliary source electrode **21**, the auxiliary drain electrode **31**, and the substrate **10** by a deposition process such as chemical vapor deposition. The buffer layer **11** is made of silicon oxide (SiOx), silicon nitride (SiNx), or a stack of silicon oxide and silicon nitride.

(48) Then, the source electrode **20** and the drain electrode **30** are formed on the buffer layer **11**, and there is the first space between the source electrode **20** and the drain electrode **30**, as shown in FIG. **7**. The source electrode **20** and the drain electrode **30** are made of a ITO, IZO, AnCl, or other metal oxide with low resistivity. For example, the source electrode **20** and the drain electrode **30** are made of a metal oxide such as ITO. A difference in an etching size of a metal oxide such as ITO in a yellow light process is small. Due to a limitation of an exposure machine, a length of the first space between the source electrode **20** and the drain electrode **30** is made smaller, so that the active layer **40** has one channel **41** with a smaller length, thereby improving the electron mobility of the channel **41** of the active layer **40**.

(49) **S303**: forming an active layer **40** on the substrate **10**, the source electrode **20**, and the drain electrode **30**, wherein the active layer **40** comprises a channel **41** and a source region **42** and a drain region **43** on both sides of the channel **41**, the source region **42** covers at least part of the source electrode **20**, the drain region **43** covers at least part of the drain electrode **30**, and the channel **41** corresponds to the first space.

(50) Specifically, a metal oxide semiconductor thin film is plated on the substrate **10**, the source electrode **20** and the drain electrode **30**, and the metal oxide semiconductor thin film is patterned to form the active layer **40**. The active layer **40** comprises the channel **41** and the source region **42** and the drain region **43** on both sides of the channel **41**, the source region **42** covers at least part of the source electrode **20**, the drain region **43** covers at least part of the drain electrode **30**, and the channel **41** corresponds to the first space, as shown in FIG. **8**.

(51) Optionally, the active layer **40** is made of a semiconductor metal oxide such as IGZO, IGZTO, IGTO, IGO, or AZTO.

(52) **S304**: forming a gate electrode **50** on a side of the active layer **40** away from the substrate **10**, wherein an orthographic projection of the gate electrode **50** on the substrate **10** covers an orthographic projection of the channel **41** of the active layer **40** on the substrate **10**, covers an orthographic projection of at least part of the source region **42** of the active layer **40** on the substrate **10**, and covers an orthographic projection of at least part of the drain region **43** of the active layer **40** on the substrate **10**.

(53) Specifically, an inorganic thin film is deposited on the active layer **40**, the source electrode **20**, the drain electrode **30**, and the buffer layer **11** by a deposition process such as chemical vapor deposition. The inorganic thin film is made of one or a combination of SiOx, SiNx, and Al₂O₃, for example, Al₂O₃/SiNx/SiOx, or SiOx/SiNx/SiOx.

(54) Next, a second metal thin film is deposited on the inorganic thin film by a deposition process such as physical vapor deposition, and the second metal thin film is patterned to form the gate electrode **50**. The gate electrode **50** is disposed corresponding to the active layer **40**. Optionally, a material of the second metal thin film may be same as a material of the first metal thin film.

(55) Then, the inorganic thin film is patterned by a self-alignment process with the gate electrode

50 as a mask to form a gate insulating layer **12**, as shown in FIG. **9**.

(56) The method for fabricating the driving substrate further comprises the following step:

(57) forming a conductive layer **60** on a side of the gate electrode **50** away from the active layer **40**, wherein the conductive layer **60** comprises a first electrode **61** and a connecting wire **62** that are spaced apart, the first electrode **61** is electrically connected to the source electrode **20** and the auxiliary source electrode **21**, and the connecting wire **62** is electrically connected to the drain electrode **30** and the auxiliary drain electrode **31**.

(58) Specifically, a passivation layer **13** is deposited on the gate electrode **50**, the source electrode **20**, the drain electrode **30**, and the buffer layer **11** by a deposition process such as chemical vapor deposition. The passivation layer **13** is made of silicon oxide (SiO_x), silicon nitride (SiN_x), or a stack of silicon oxide and silicon nitride.

(59) The passivation layer **13** is patterned to form a plurality of first via holes **131** and a plurality of second via holes **132** in the passivation layer **13**, as shown in FIG. **10**. The first via holes **131** penetrate through the passivation layer **13**, one of the first via holes **131** exposes another part of the source electrode **20**, and the other one of the first via holes **131** exposes another part of the drain electrode **30**. The second via holes **132** penetrate through the passivation layer **13** and the buffer layer **11**, one of the second via holes **132** exposes a part of the auxiliary source electrode **21**, and the other one of the second via holes **132** exposes a part of the auxiliary drain electrode **31**.

(60) Next, a conductive layer **60** is formed on the passivation layer **13**, and the conductive layer **60** is patterned to form a first electrode **61** and a connecting wire **62**, as shown in FIG. **1**. The first electrode **61** is electrically connected to the source electrode **20** through one first via hole **131**. At the same time, the first electrode **61** is also electrically connected to the auxiliary source electrode **21** through one second via hole **132**. The connecting wire **62** is electrically connected to the drain electrode **30** through the other first via hole **131**. At the same time, the connecting wire **62** is also electrically connected to the auxiliary drain electrode **31** through the other second via hole **132**. In this way, a mask can realize the electrical connection between the source electrode **20** and the auxiliary source electrode **21**, the electrical connection between the drain electrode **30** and the auxiliary drain electrode **31**, and the electrical connection between the first electrode **61** and the source electrode **20**, which is beneficial to save cost.

(61) Optionally, the conductive layer **60** is made of a metal oxide electrode material such as IZO and ITO, or may be a three-metal layer anti-reflection structure of IZO/Mo/Cu, so as to reduce an influence of a light emitting unit on the channel **41** of the active layer **40**.

(62) Based on a same inventive concept, the present disclosure further provides a display panel. The display panel comprises the driving substrate of one of the foregoing embodiments. The display panel may be a liquid crystal display panel, an OLED display panel, an LED display panel, or the like. For example, when the display panel is an LED display panel, the display panel further comprises an LED chip, and the LED chip is bound on the driving substrate **100**. Specifically, the driving substrate **100** further comprises a second electrode disposed on a same layer as the first electrode **61**. The LED chip is electrically connected to the first electrode **61** and the second electrode of the driving substrate **100** to bound the LED chip to the driving substrate **100**.

(63) According to aforementioned embodiments, the following can be known.

(64) The present disclosure provides a driving substrate, a method for fabricating the same, and a display panel. The driving substrate comprises a substrate, a source electrode, a drain electrode, and an active layer. The source electrode, the drain electrode, and the active layer are disposed on the substrate. There is a first space between the source electrode and the drain electrode. The active layer is further disposed on the source electrode and the drain electrode. The active layer comprises a channel and a source region and a drain region on both sides of the channel. The source region covers at least part of the source electrode. The drain region covers at least part of the drain electrode. The channel corresponds to the first space. In this way, a length of the channel of the active layer is defined by a space between the source electrode and the drain electrode, so as to

reduce the length of the channel of the active layer, thereby improving an electron mobility of the channel of the active layer and solving a problem that improvement of an electron mobility of a current thin film transistor is limited.

(65) In the above embodiments, the description of each embodiment has its own emphasis. For parts not detailed in one embodiment, reference may be made to the related descriptions in other embodiments.

(66) The embodiments of the present disclosure are described in detail above. The present disclosure uses specific examples to describe principles and embodiments of the present application. The above description of the embodiments is only for helping to understand the technical solutions of the present disclosure and its core ideas. It should be understood by those skilled in the art that they can modify the technical solutions recited in the foregoing embodiments, or replace some of technical features in the foregoing embodiments with equivalents. These modifications or replacements do not cause essence of corresponding technical solutions to depart from the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. A driving substrate, comprising: a substrate; a source electrode and a drain electrode disposed on the substrate, wherein there is a first space between the source electrode and the drain electrode; an active layer disposed on the substrate, the source electrode, and the drain electrode, and comprising a channel and a source region and a drain region on both sides of the channel, wherein the source region covers at least part of the source electrode, the drain region covers at least part of the drain electrode, and the channel corresponds to the first space; a gate electrode disposed on a side of the active layer away from the source electrode and the drain electrode, and disposed corresponding to the active layer, wherein an orthographic projection of the gate electrode on the substrate covers an orthographic projection of the channel of the active layer on the substrate, covers an orthographic projection of at least part of the source region of the active layer on the substrate, and covers an orthographic projection of at least part of the drain region of the active layer on the substrate; an auxiliary source electrode disposed between the source electrode and the substrate and electrically connected to the source electrode; and an auxiliary drain electrode disposed between the drain electrode and the substrate and electrically connected to the drain electrode.
2. The driving substrate according to claim 1, further comprising: a conductive layer disposed on a side of the gate electrode away from the active layer, insulated from the gate electrode, and comprising a first electrode and a connecting wire that are spaced apart, wherein the first electrode is electrically connected to the source electrode and the auxiliary source electrode, and the connecting wire is electrically connected to the drain electrode and the auxiliary drain electrode.
3. The driving substrate according to claim 2, wherein an orthographic projection of the first electrode on the substrate at least covers the orthographic projection of the channel on the substrate.
4. The driving substrate according to claim 1, further comprising: a buffer layer covering the auxiliary source electrode and the auxiliary drain electrode, wherein the source electrode and the drain electrode are disposed on the buffer layer, and are respectively electrically connected to the corresponding source auxiliary electrode and the drain auxiliary electrode through via holes of the buffer layer.
5. The driving substrate according to claim 1, wherein in a direction perpendicular to the substrate, a thickness of the source electrode is less than a thickness of the auxiliary source electrode, and a thickness of the drain electrode is less than a thickness of the auxiliary drain electrode.
6. The driving substrate according to claim 1, wherein an orthographic projection of the auxiliary source electrode on the substrate at least covers the orthographic projection of the channel on the substrate.
7. The driving substrate according to claim 1, wherein the source electrode and the drain electrode

are made of metal oxide.

8. A display panel, comprising a driving substrate, wherein the driving substrate comprises: a substrate; a source electrode and a drain electrode disposed on the substrate, wherein there is a first space between the source electrode and the drain electrode; an active layer disposed on the substrate, the source electrode, and the drain electrode, and comprising a channel and a source region and a drain region on both sides of the channel, wherein the source region covers at least part of the source electrode, the drain region covers at least part of the drain electrode, and the channel corresponds to the first space; a gate electrode disposed on a side of the active layer away from the source electrode and the drain electrode, and disposed corresponding to the active layer, wherein an orthographic projection of the gate electrode on the substrate covers an orthographic projection of the channel of the active layer on the substrate, covers an orthographic projection of at least part of the source region of the active layer on the substrate, and covers an orthographic projection of at least part of the drain region of the active layer on the substrate; an auxiliary source electrode disposed between the source electrode and the substrate and electrically connected to the source electrode; and an auxiliary drain electrode disposed between the drain electrode and the substrate and electrically connected to the drain electrode.

9. The display panel according to claim 8, wherein the driving substrate further comprises: a conductive layer disposed on a side of the gate electrode away from the active layer, insulated from the gate electrode, and comprising a first electrode and a connecting wire that are spaced apart, wherein the first electrode is electrically connected to the source electrode and the auxiliary source electrode, and the connecting wire is electrically connected to the drain electrode and the auxiliary drain electrode.

10. The display panel according to claim 9, wherein an orthographic projection of the first electrode on the substrate at least covers the orthographic projection of the channel on the substrate.

11. The display panel according to claim 10, wherein the driving substrate further comprises: a buffer layer covering the auxiliary source electrode and the auxiliary drain electrode, wherein the source electrode and the drain electrode are disposed on the buffer layer, and are respectively electrically connected to the corresponding source auxiliary electrode and the drain auxiliary electrode through via holes of the buffer layer.

12. The display panel according to claim 8, wherein in a direction perpendicular to the substrate, a thickness of the source electrode is less than a thickness of the auxiliary source electrode, and a thickness of the drain electrode is less than a thickness of the auxiliary drain electrode.

13. The display panel according to claim 8, wherein an orthographic projection of the auxiliary source electrode on the substrate at least covers the orthographic projection of the channel on the substrate.

14. The display panel according to claim 8, wherein the source electrode and the drain electrode are made of metal oxide.

15. A method for fabricating a driving substrate, comprising: providing a substrate; forming a first metal thin film on the substrate by physical vapor deposition; patterning the first metal thin film to form an auxiliary source electrode and an auxiliary drain electrode; depositing a buffer layer on the auxiliary source electrode, the auxiliary drain electrode, and the substrate by chemical vapor deposition; forming a source electrode and a drain electrode on the buffer layer, wherein there is a first space between the source electrode and the drain electrode; forming an active layer on the substrate, the source electrode, and the drain electrode, wherein the active layer comprises a channel and a source region and a drain region on both sides of the channel, the source region covers at least part of the source electrode, the drain region covers at least part of the drain electrode, and the channel corresponds to the first space; and forming a gate electrode on a side of the active layer away from the substrate, wherein an orthographic projection of the gate electrode on the substrate covers an orthographic projection of the channel of the active layer on the substrate, covers an orthographic projection of at least part of the source region of the active layer

on the substrate, and covers an orthographic projection of at least part of the drain region of the active layer on the substrate.

16. The method for fabricating the driving substrate according to claim 7, further comprising: forming a conductive layer on a side of the gate electrode away from the active layer, wherein the conductive layer comprises a first electrode and a connecting wire that are spaced apart, the first electrode is electrically connected to the source electrode and the auxiliary source electrode, and the connecting wire is electrically connected to the drain electrode and the auxiliary drain electrode.
